

Stefan Vilceanu

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EDUCATION

- RWTH Aachen University - M.Sc. Computer Engineering** Apr. 2023 – Mar. 2025
Machine Learning, Embedded Systems, Computer Networks Aachen, DE
Thesis: Uncertainty Estimates for Transformer-based Radar-, Camera-, and LiDAR-driven 3D Object Detectors
- RWTH Aachen University - B.Sc. Electrical Engineering** Oct. 2019 – Mar. 2023
Electrical Circuits, Telecommunications, Embedded Systems Aachen, DE

EXPERIENCE

- Porsche Engineering** Apr. 2025 – Present
Software Engineer Stuttgart, DE & Timisoara, RO
- Designed and implemented an AI Automation Agent for testing the UI of Porsche's Infotainment Systems
 - Prototyped a real-time Speech AI Agent for testing the Digital Assistant of Porsche's Infotainment Systems
 - Improved the existing testing framework by creating an AI Resolver to identify and repair broken testcases
- ETAS GmbH** Dec. 2024 – Apr. 2025
Software Engineer Intern Stuttgart, DE
- Analyzed real-world cyberattacks on vehicles and explored how existing SIEM systems could detect them
 - Built a prototype transformer-based language model to generate accurate security queries from natural language
- IKA RWTH Aachen** Apr. 2024 – Nov. 2024
Computer Vision Research Assistant Aachen, DE
- Developed a Lane Analysis Algorithm using YoloV2, Inverse Perspective Mapping and Bayesian Filtering
 - Implemented ROS2 nodes to enable algorithm deployment on a real testing truck
- WZL RWTH Aachen** Mar. 2023 – Apr. 2024
Robotics Research Assistant Aachen, DE
- Developed and trained deep learning models to predict torque in robot joints
 - Implemented MATLAB scripts to generate robot trajectories for robot experiments on the MABI 100
- IAEW RWTH Aachen** Jun. 2021 – Jul. 2022
Smart Grids Research Assistant Aachen, DE
- Developed a Python framework to visualize hybrid electrical grids using OpenStreetMap
 - Designed the planning of hybrid AC/DC grids using the Genetic algorithm

PROJECTS

- Inference Pipeline for an Autonomous Car** | C++, RPi5, HailoRT, YOLOv8, Depth Anything V2
- Real-time C++ perception-to-actuation pipeline on RPi 5 + Hailo-8L, using YOLOv8 and Depth Anything V2
 - Implemented person-following autonomy using SORT tracking with PD/PI control from depth data
 - Added hands-free voice control via custom ONNX wake-word models streamed into the pipeline
- Autonomous FPV Drone** | Betaflight, ArduPilot, ELRS, Flight Automation, Flight Controller, Waypoint Navigation
- Built an FPV drone from scratch, flashing and configuring ArduPilot, ELRS and the video system
 - Tuned and improved flight performance through blackbox data logging and PID analysis in PIDtoolbox
 - Configured GPS-based autonomous waypoint navigation using ArduPilot, enabling preplanned missions
- Autonomous UAV Simulation – Ongoing** | Jetson Orin, Gazebo, QGC, ROS2, PX4, Docker, TensorRT, VSLAM
- Architecting an autonomous UAV stack split across a WSL workstation and a Jetson Orin Nano
 - Building a six-phase perception and guidance pipeline including TensorRT-accelerated detection, Isaac ROS cuVSLAM visual-inertial odometry and min-jerk trajectory generation with L1 guidance

TECHNICAL SKILLS

Languages: C++, Python, C, Bash/Shell
Robotics & Autonomy: ROS2, PX4, Betaflight, ArduPilot, Gazebo, VSLAM, PID Control, Estimation, Filtering
AI & Computer Vision: PyTorch, OpenCV, Scikit-learn, HF Transformers, LangChain/Graph
Edge AI & Deployment: NVIDIA Jetson, TensorRT, ONNX, HailoRT, Docker, FastAPI
Tools & Systems: Linux, WSL, Git, GitLab CI/CD, Nginx, Weights & Biases